Hashir Ahmed Buriro NAVTTC-AI (ML & DL) Date: 10/May/2026

Final Project

FINAL PROJECT REPORT

Classification using Machine Learning and Deep Learning Models

Introduction

Machine Learning and Deep Learning are widely used techniques for solving real-world classification problems. These models learn patterns from data and make predictions based on input features.

In this project, we implement a complete classification pipeline using multiple Machine Learning and Deep Learning algorithms. The goal is to analyze data, train different models, and compare their performance to identify the best-performing approach.

This project also demonstrates the importance of data preprocessing, feature engineering, and model evaluation in building an effective predictive system.

Objective

The objective of this project is to build a classification system that predicts whether an individual's income is above or below a certain threshold using demographic and employment-related data.

The project also compares multiple models to determine the most accurate and efficient algorithm.

Dataset Description

Dataset Name:

Adult Income Dataset (Best Dataset for Classification Tasks)

Source:

Publicly available dataset used in ML classification problems

Problem Type:

Binary Classification

Target Variable:

Income Class (≤50K or >50K)

Features:

- Age
- Workclass
- Education
- Marital Status
- Occupation
- Relationship
- Race
- Gender
- Hours-per-week
- Native Country

Exploratory Data Analysis (EDA)

Steps:

- Missing values handling
- Outlier detection
- Feature distribution analysis
- Correlation analysis

Visualizations:

- Distribution plots
- Boxplots
- Heatmap
- Pairplots

Key Insights:

- Education strongly affects income
- Working hours are important predictors
- Data contains nonlinear patterns

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Data Preprocessing

- Missing value handling
- Encoding categorical variables
- Feature scaling
- Train-test split

Models Implemented

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Support Vector Machine (SVM)
5. Neural Network (MLP)

Model Evaluation

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix

Results Summary

- 🎄 Random Forest → **Best performance**
- 📈 SVM → Strong but slow
- 🧠 Neural Network → Good after tuning
- 🌳 Decision Tree → Overfitting issue
- 🔴 Logistic Regression → Lowest performance

Analysis & Discussion

- Random Forest performed best due to ensemble learning
- Decision Tree showed overfitting
- Random Forest reduced overfitting
- Linear models failed to capture complex patterns
- Deep learning performed well with tuning

Future Improvements

- Hyperparameter tuning
- Cross-validation
- Advanced deep learning models
- Feature engineering
- Deployment using Flask / Streamlet

Conclusion

This project successfully implemented a complete machine learning pipeline for classification using both classical Machine Learning and Deep Learning models. The workflow included data preprocessing, exploratory data analysis, model training, and performance evaluation. Multiple models such as Logistic Regression, Decision Tree, Random Forest, SVM, and Neural Network were trained and compared on the same dataset to ensure fair evaluation.

The results show that Random Forest performed the best due to its ensemble learning approach, which improves accuracy and reduces overfitting. Overall, the project highlights that ensemble methods and nonlinear models perform better on complex datasets, while proper preprocessing and model selection play a critical role in achieving high performance.

Final Statement

This project demonstrates a complete machine learning workflow including data preprocessing, model training, evaluation, and comparison. It highlights that ensemble methods like Random Forest provide the best performance for this classification problem.